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Palaeomagnetism and K-Ar ages of Neogene rocks of northern Taiwan: tectonics of the arc junction of Ryukyu and Luzon arcs

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SUMMARY

More than 100 block samples were collected for palaeomagnetic and geochronological studies from 14 sites in Miocene and Plio–Pleistocene volcanic rocks in the western foothills province of northern Taiwan, in an attempt to understand the tectonic evolution of the junction of the Ryukyu and the Luzon arcs. Potassium-argon dating yields ages of 7.9 ± 0.5 Ma and 10.1 ± 0.6 Ma for two samples from the Miocene Chiopanshan volcanic series, and 1.2 ± 0.1 Ma for a sample from the Plio–Pleistocene Chilung volcanic group. The palaeomagnetic direction from seven sites of the Miocene volcanics shows no deflection of declination value ($D = 5.9°$, $I = 43.8°$, $\alpha_{95} = 14.0°$). The direction from the volcanic rock of 1.2 Ma is $D = 2.5°$ and $I = 53.3°$. The reliability of the direction is ascertained through thermal demagnetization, positive fold test and the presence of normal and reversed polarities. Comparison of Neogene palaeopoles between northern Taiwan and China indicates that little relative tectonic movement has occurred in northern Taiwan since 20 Ma. These results establish that the arcuate shape of geological belts in northern Taiwan is not due to tectonic bending but to an inherited geometry. Reconstruction of the 10 Ma palaeoposition of Taiwan and the south Ryukyu arc implies that the cusp shape at the arc junction has already existed prior to 10 Ma.

Key words: arc junction, cusp shape, K-Ar ages, northern Taiwan, palaeomagnetism.

INTRODUCTION

The island chain along the western margin of the Philippine sea is a series of arcs and cusps extending from the southwest Japan arc to the Mindanao–Halmahera arc. Taiwan lies on the junction of the northeast–southwest-trending Ryukyu arc and the north–south-trending Taiwan–Luzon arc. The junction at northern Taiwan forms a cusp where the included angle of the two arcs is smaller than 100°.

The structural pattern in northern Taiwan is characterized by a bent feature which is convex to the northwest (Fig. 1; Ho 1988). The rock formations in Taiwan occur in long narrow belts parallel to the NNE-trending long axis of the island, which turns NE–SW in northern Taiwan and forms a spectacular bend. Major faults and fold axes follow this bent structural pattern. The shape of the bend looks to be a part of the cusp at the arc junction. The structural bend must be a clue to the formation of the arc–arc junction.

In this paper, we will attempt to give a further contribution to the interpretation of the island arc junction and their cusp feature using palaeomagnetic and geochronological methods. Since palaeomagnetic study is useful for understanding the kinematic history of island arcs and their junctions, it gives information about the origin of the cusp. Miocene rocks were chosen, because many tectonic events have occurred since the Miocene: the south Ryukyu rotated clockwise later than 10 Ma (Miki, Matsuda & Otofuji 1990), and the Luzon arc began to collide with the Eurasian Plate in 7–8 Ma (Pelletier & Stephan 1986; Stephan *et al.* 1986; Jahn *et al.* 1986) or 4–5 Ma (Chi,

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Namson & Suppe 1981; Barrier & Angelier 1986; Teng 1987; Teng 1990).

GEOLOGY AND SAMPLING

The geological belts in Taiwan become progressively younger from the eastern Central Range westward to the coastal plain (Fig. 1; Ho 1986, 1988). The western foothills province contains late Oligocene, through early Pleistocene continental sedimentary formations. The Plio–Pleistocene orogeny deformed all the sedimentary units and formed a non-metamorphic fold-and-thrust belt in the western foothills (Ho 1986, 1988). The Chuchih fault represents a large fault that separates the western foothills province from the Eocene through Miocene low-grade metamorphic terrain (western Central Range) to the east.

Volcanic activity occurred intermittently during the Miocene, and produced basaltic tuffs and lava flows in the northern part of the western foothills. Three volcanic stages have been identified in the Miocene strata: the Kungkuan, the Chienshih and the Chiopanshan, which roughly correspond to early Miocene, middle Miocene and late Miocene, respectively (Ho 1988). The intercalated sedimentary strata contain nannofossil zones NN1 to NN11 and planktonic foraminifera zones N5 and N16 (Huang & Huang 1984), suggesting an interval from 23 Ma to 8 Ma.

Samples of tuffs and basaltic lava flows were collected from thirteen sites in the Miocene strata (Fig. 1): eight from the Kungkuan stage and five from the Chiopanshan stage. The tilt for the tectonic correction was estimated from the intercalated sedimentary strata.

The geological thrust belts have an arcuate shape convex

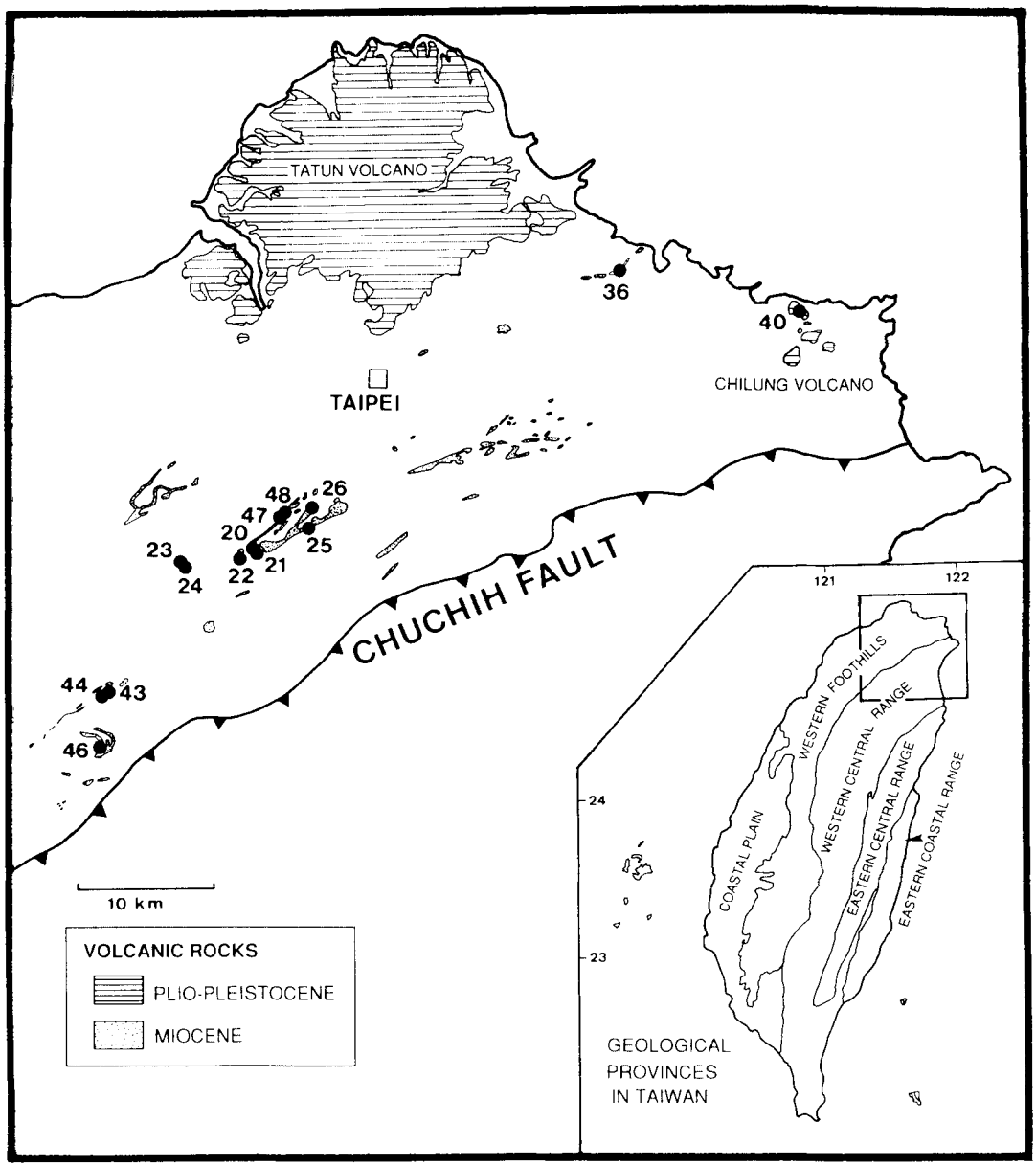


Figure 1. Geological map of sampling localities showing the distribution of Miocene and Plio–Pleistocene volcanic rocks. Solid circles mark sampling localities. Major geological provinces are also shown (after Ho 1988).

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to the west. The bend occurred at about 24°30′N: The strike of the structure is nearly N10°E to the south of 24°30′N, and changing to N65°E at the northeastern end of Taiwan. Almost all of the Miocene volcanics involving our sampling sites distribute to the northeast of the bend (Fig. 1). The directions of faults and fold axes are ENE–WSW in the sampling area. The strikes of the bedding in sampling sites trend about N60°E and parallel to the major structure of faults and folds.

Volcanic activity began again in the Plio–Pleistocene. Successive eruptions of two andesitic and dacitic volcanoes formed the Tatun Volcanic Group in the west and the Chilung Volcanic Group in the east (Fig. 1). We also sampled dacite lavas at one site in the Chilung Volcanic Group.

Block samples were taken and a magnetic compass was used for taking orientation data. Each site typically comprised 10 independently oriented samples, distributed over a distance ranging up to 30 m.

GEOCHRONOLOGY

K-Ar whole-rock dating was conducted on two samples from the Miocene volcanic rocks of the Chiopanshan stage (site NT24 and NT43) and one sample from the Chilung Volcanic Group (site NT40). The samples from the Miocene volcanics are dolerites with an ophitic texture, and the sample from the Chilung Volcanic Group is porphyritic dacite. Thin sections of the samples were examined to determine their suitability for whole rock K-Ar dating (see appendix), according to the criteria discussed by Mankinen & Dalrymple (1972).

Whole-rock samples were prepared for argon analysis by crushing to 80–100 mesh size. A portion of each sample was further ground in an agate mortar and used for potassium analysis. Details of the analytical procedures for potassium and argon isotope determinations have been reported by Itaya & Takasugi (1988). Ages and errors were calculated with the method by Nagao & Itaya (1988), using the decay constants determined by Steiger & Jäger (1977).

The Miocene volcanic rocks from the Chiopanshan stage have ages of 7.9 ± 0.5 Ma and 10.1 ± 0.6 Ma (Table 1). The age for the Chilung dacite is 1.2 ± 0.1 Ma. The dacite age falls in the period between 0.2 Ma and 1.7 Ma, which has been radiometrically established for the Chilung volcanic group by Juang & Bellon (1984).

Table 1. Results of K-Ar dating on dolerites and dacite from northern Taiwan. The constants used in the calculation are $\lambda_\beta = 4.962 \times 10^{-10}\ \mathrm{yr}^{-1}$, $\lambda_e = 0.581 \times 10^{-10}\ \mathrm{yr}^{-1}$, and $^{40}\mathrm{K} = 1.167 \times 10^{-4}$ atom per atom of natural potassium (Steiger & Jäger 1977).

Site No.	Locality	Sample	K (wt%)	$^{40}Ar_{rad}$ (10^{-8}cc/g)	Age (Ma)	Ar_{air}/Ar_{total} (%)
Chiopanshan stage:						
NT24	121°24′5″ E 24°56′34″ N	Dolerite	0.580	17.8±0.87	7.91±0.45	71.9
NT43	121°20′35″ E 24°51′33″ N	Dolerite	0.426	16.8±0.58	10.1±0.62	64.0
Chilung volcano:						
NT40	121°50′22″ E 25°7′10″ N	Dacite	1.511	7.06±0.40	1.20±0.07	76.4

PALAEOMAGNETISM

A total of 109 block samples were taken for the palaeomagnetic study. More than two cylindrical specimens 2.5 cm length and 2.5 cm diameter were obtained from each sample.

Natural remanent magnetizations (NRMs) were measured using a two-axis ScT cryogenic magnetometer or a spinner magnetometer according to the intensity of magnetization. The NRM intensities were fairly high, typically of the order of $1\ \mathrm{A\ m^{-1}}$.

Two or three samples were chosen for a pilot study. One specimen from each pilot sample was subjected to stepwise thermal demagnetization up to 600 °C. Other specimens were subjected to stepwise alternating field (AF) demagnetization, up to a maximum field of 100 mT.

Samples from eight sites showed stable magnetic components during progressive demagnetization (Fig. 2). All of these samples have a high temperature component with unblocking temperature between 500 °C and 600 °C, indicating the magnetic carrier of magnetite. The high temperature component in some samples is masked by a low temperature component which is removed at 200 °C (Fig. 2c). We regarded the direction of the high temperature component as the characteristic direction for each specimen. This characteristic direction could also be distinguished by the AF demagnetization technique: a direction similar to the high temperature component was retrieved after demagnetization in a peak field of 5 mT (Fig. 2). The median destructive field (MDF) ranges between 10 mT and 40 mT. Because the thermal demagnetization method appeared to be more effective for removing magnetic overprint, the remaining specimens were treated by the thermal method.

The characteristic direction of each specimen from the eight sites was determined by principal component analysis (Kirschvink 1980) using more than five points on the stepwise thermal demagnetization diagram. Most of the specimens have maximum angular dispersion (MAD) values less than 5°, indicating the good linearity of the demagnetization paths. Only characteristic directions with MAD values less than 10° were used for the site-mean calculations (Table 2).

Samples from remaining four sites have no stable components or NRMs. The demagnetization path is rather erratic (MAD > 10°). The samples from these sites were excluded from further work in this study.

Miocene volcanics

The reliable characteristic directions were obtained from seven sites (Fig. 3). Various lines of evidence indicate that the characteristic directions from seven sites are primary origin. (1) The characteristic directions provide a positive fold test which is significant at 95 per cent confidence level (McElhinny 1964): k ratio increases from $k_1 = 5.7$ to $k_2 = 19.4$ ($k_2/k_1 = 3.4$) for seven sites (Fig. 3). (2) Both normal and reversed polarities are observed and these directions are mutually antipodal. (3) The NRMs reside in magnetite. Chemical remanent magnetization as a secondary magnetization in magnetite is uncommon (McCabe *et al.* 1983). The primary magnetite carries the characteristic directions.

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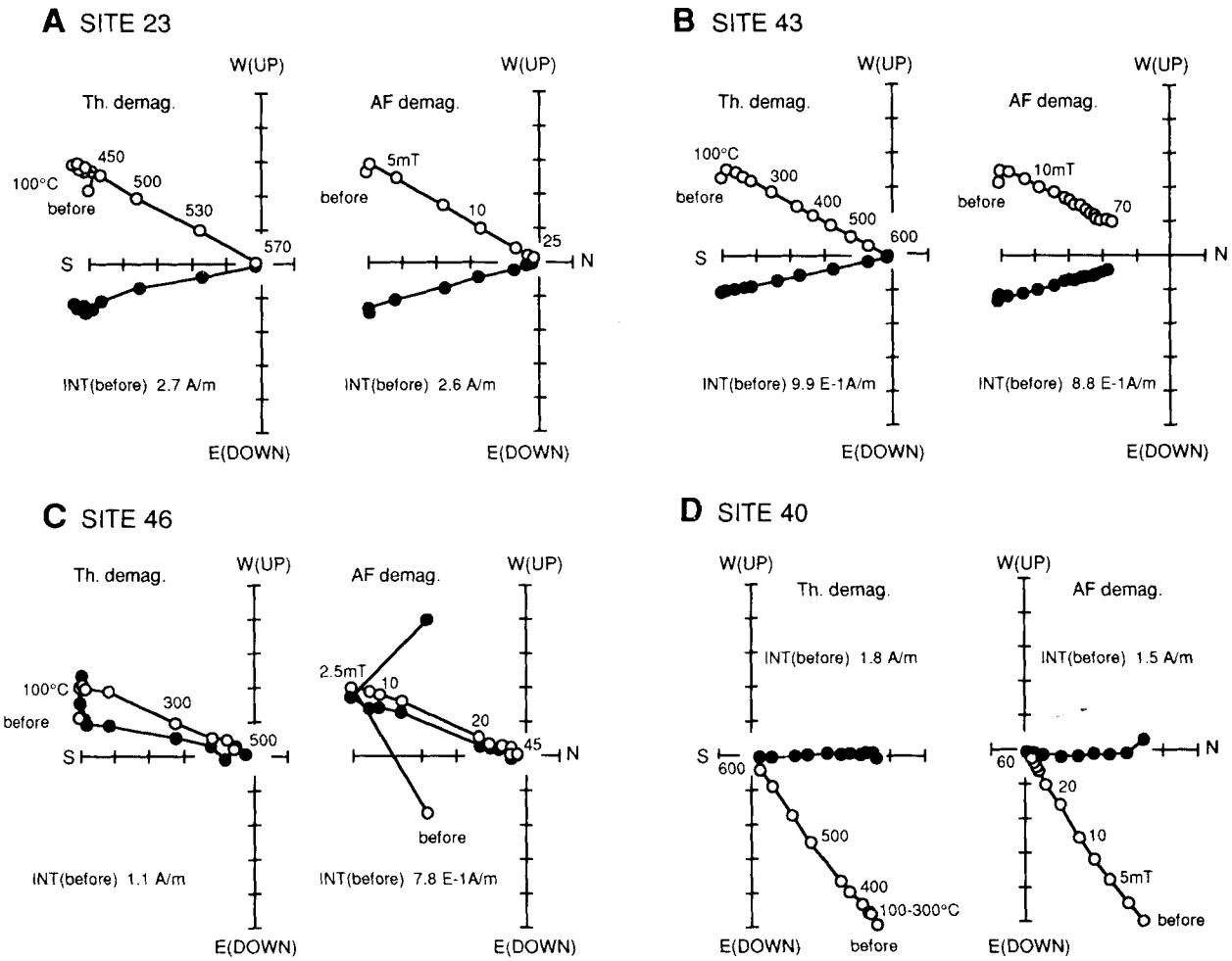


Figure 2. Orthogonal projection plots for NRM stability examinations. Typical examples of comparison between behaviours during progressive thermal (Th. demag.) and AF demagnetization (AF demag.) for pilot specimens. A–C, examples for Miocene volcanic rocks (A, tuff; B and C, basalt). D, example for the Chilung Volcanic Group. Open (solid) circles are on the vertical (horizontal) plane.

Table 2. Summary of palaeomagnetic results for northern Taiwan. N—number of rock samples; Dec—declination, Inc—inclination, uncorrected/corrected for bedding tilt; α_{95}—radius of 95 per cent confidence circle; k—precision parameter; Plat—latitude, and Plong—longitude of site-mean pole (after tilt correction). *—Pole calculated from the direction before tilt correction.

			Uncorrected		Corrected						
Geological sequence	Site	N	Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)	k	α_{95} (°)	Plat. (°N)	Plong. (°E)	Rock type
Miocene:											
Kungkuan volcanics	NT25	7	2.0	11.8	15.4	42.5	113.4	5.7	76.0	209.8	Tuff
	NT47	4	-89.2	-53.2	-178.4	-49.2	73.0	10.8	84.7	136.7	Tuff
Chiopanshan volcanics	NT23	7	167.3	-24.2	166.0	-36.0	202.8	4.3	76.1	13.0	Tuff
	NT24	9	-169.9	-42.5	-145.2	-52.7	185.1	3.8	58.7	188.0	Dolerite
	NT43	7	164.5	-26.4	167.5	-60.3	542.6	2.6	70.7	92.0	Dolerite
	NT44	7	174.3	-8.6	178.8	-41.4	323.2	3.4	88.4	343.8	Dolerite
	NT46	10	-163.2	10.7	-164.4	-16.5	60.6	6.3	67.7	256.9	Basalt
Plio-Pleistocene:											
Chilung Volcanics	NT40	9	2.5	53.3	273.4	3.1			*80.9	*134.7	Dacite

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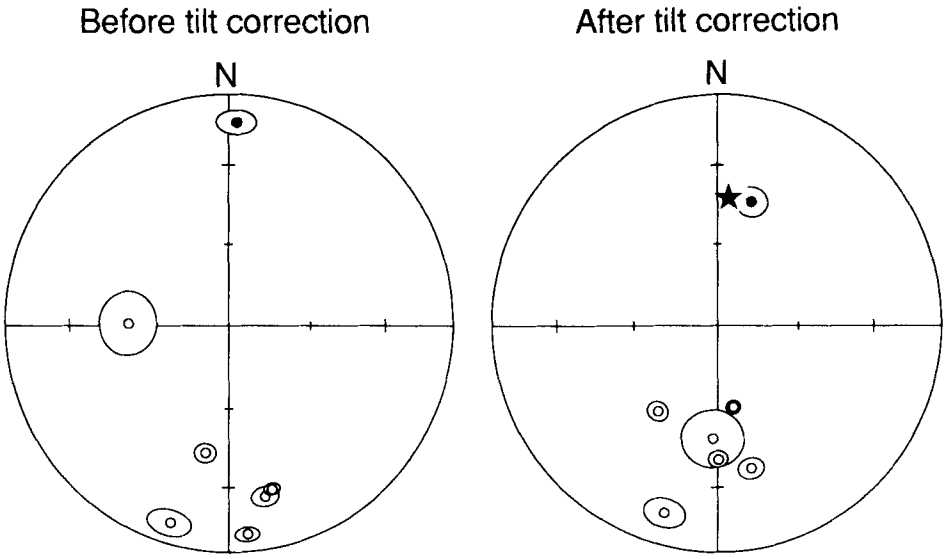


Figure 3. Palaeomagnetic site mean directions with 95 per cent confidence circles from Miocene volcanics. Comparison between directions before and after tilt corrections. Star shows the mean direction for the seven sites.

(4) Little alteration is observed in the minerals under the microscopic observation.

The palaeomagnetic results are obtained from two sites of early Miocene (Kungkuan stage) and five sites of late Miocene (Chiopanshan stage). The palaeomagnetic directions of early Miocene are not significantly different from those of late Miocene. The mean direction of the characteristic directions after tilt correction ($D = 5.9°$, $I = 43.8°$, $\alpha_{95} = 14.0°$) is accepted to be a primary palaeomagnetic direction of Miocene in northern Taiwan.

Chilung volcanics

The characteristic direction from one site is $D = 2.5°$, $I = 53.3°$ (Fig. 3). This direction probably represents primary magnetization based on the lack of alteration of minerals (the thin section was examined for K-Ar dating), the stability of magnetization, and the high unblocking temperature of 600 °C.

DISCUSSION

The obtained K-Ar ages of the Chiopanshan volcanic rocks are in good agreement with those based on palaeontological studies. The volcanic rocks of the Chiopanshan stage are intercalated with the strata of the Nanchuang Formation (Ho 1988). This formation belongs to the N14 to N16 planktonic foraminiferal zones defined by Blow (1969) and to the NN8 to NN10 calcareous nannofossil zones by Martini (1971): (Huang & Huang 1984). These zones are assigned to the interval between 13 Ma and 8 Ma. Thus, our K-Ar ages of 7.9 Ma and 10.1 Ma are compatible with palaeontological data. The geochronological results confirm the hypothesis that the volcanism occurred contemporaneously with the deposition of the Miocene strata (Ho 1988).

Palaeomagnetic analyses indicate that our non-deflected magnetic direction from the Miocene volcanic rocks represents a reliable and realistic palaeomagnetic direction of Miocene age. The geomagnetic secular variation is averaged out in the mean calculation for the palaeomagnetic direction, because our directions are obtained from rocks with ages from early to late Miocene, covering at least two polarity epochs of normal and reversed.

In northern Taiwan, Hsu *et al.* (1966) observed northerly palaeomagnetic directions in the basaltic lavas of the Chiopanshan stage, and Lee *et al.* (1991) reported southerly directions with small clockwise deflection from the Miocene sedimentary rocks. The former data were obtained without thermal demagnetization and principal component analysis. In the latter study, detailed information concerning to the thermal of AF demagnetization experiments on the magnetic stability was not provided. These data passed neither the reversal test nor the fold test. We therefore suggest that these data are less reliable than our data and accordingly we did not combine the palaeomagnetic data of Hsu *et al.* (1966) and Lee *et al.* (1991) with our new data.

We compared the palaeomagnetic pole position obtained from northern Taiwan in this study with the Neogene pole position for the Asian continent (Fig. 4). Because the amalgamation of almost all tectonic blocks in the Asian continent (e.g. north and south China blocks) was completed by late Jurassic (Lin, Fuller & Zhang 1985; Chan 1991; Zheng *et al.* 1991; Enkin *et al.* 1992), we chose the Neogene pole of the North China Block as a reference pole of the Asian continent. The Miocene pole position of northern Taiwan (191.1°E, 84.5°N, $\alpha_{95} = 14.0°$) is in excellent agreement with that for the Neogene North China block (192.5°E, 85.2°N) determined by Zheng *et al.* (1991) and Zhao *et al.* (1990). This comparison indicates that northern Taiwan has undergone little tectonic rotation with respect to the Asian continent since about 20 Ma in early Miocene, when the Kungkuan volcanics erupted.

This excellent agreement of the poles does not support the 'Cainozoic bending' hypothesis in northern Taiwan. Many workers have explained the particular bend of geological structure at about 24°N as the result of the

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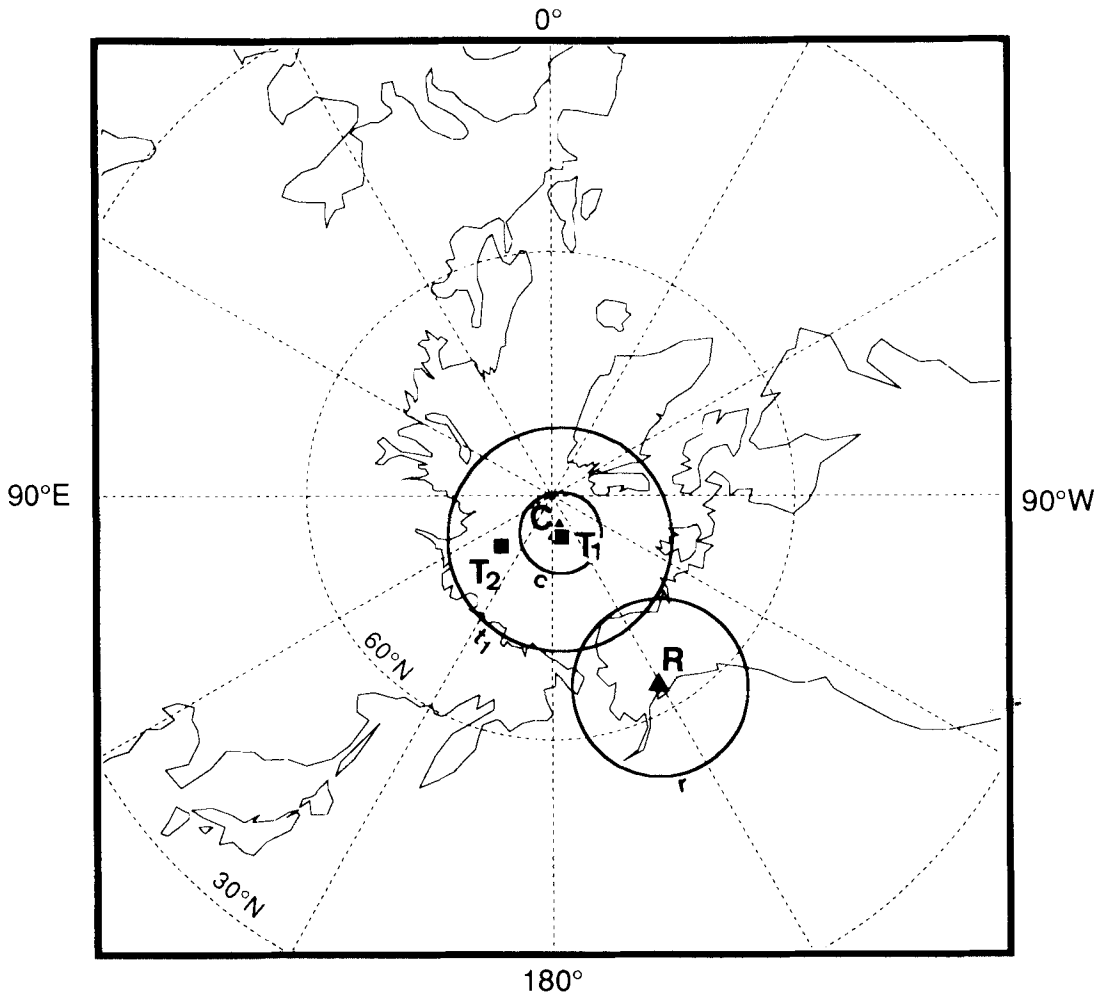


Figure 4. Stereographic projections of palaeopoles with 95 per cent confidence circles. Squares are in this study. T_1(square) with the circle t_1, pole for Miocene northern Taiwan. C(triangle) with the circle c, pole for China block compiled by Zheng *et al.* 1991. R(triangle) with the circle r, Eocene pole for the South Ryukyu arc by Miki *et al.* (1990). T_2(square), pole for a site of the Chilung Volcanic Group.

clockwise rotation of northern Taiwan in the Neogene time (Wu 1978; Angelier, Barrier & Chu 1986; Angelier *et al.* 1990b; Lee *et al.* 1991). Our palaeomagnetic results were obtained from the area between 24°49′N and 24°59′N in latitude, and between 121°20′E and 121°30′E in longitude where the strike of the structure is N60°E. Our palaeomagnetic study shows little rotation in this area of northern Taiwan. We therefore conclude that the remarkable bend represents the original shape of the continental margin controlled by older inherited structures rather than being a secondary feature formed by tectonic rotation.

The 1.2 Ma pole for the Chilung volcanic group (134.7°E, 80.8°N) is fairly close to the present geographic north pole or the Miocene pole of Taiwan (Fig. 4). The Chilung area probably has not rotated since 1.2 Ma. This result is consistent with our conclusion that little rotation has occurred in post-20 Ma northern Taiwan.

The following geological evidence in the Taiwan Strait indicates that no drifting of Taiwan from the Asian continent has occurred since Miocene: thick Palaeogene and Neogene flysch sediments in western Taiwan accumulated on the continental crust that extends into mainland China (Zhang, Liou & Coleman 1984). Seismic reflection profiles show that the rifted Mesozoic basement in the Taiwan Strait is overlain by flat-lying Cainozoic sedimentary sequences (Wageman, Hilde & Emery 1970; Juan 1986; Sun 1982; Angelier *et al.* 1990a; Teng 1990), indicating that there have been no large tectonic events in the strait since Cainozoic. Taiwan has been a part of the stable Asian continental margin. The above geological arguments combined with the palaeomagnetic evidence show both the northern and main parts of Taiwan have remained fixed with respect to the Asian continent and have undergone neither tectonic rotation nor north–south translation with respect to the Asian continent since 20 Ma.

In Fig. 4, we compare the pole positions of Taiwan and south Ryukyu. The Taiwan pole determined from this study deviates by 22° from the Eocene Ryukyu pole (209.3°E, 63.0°N) reported by Miki *et al.* (1990). This deviation is ascribable to the post-10 Ma clockwise rotation of the south Ryukyu arc associated with the back-arc opening of the Okinawa Trough (Miki *et al.* 1990). While the south Ryukyu arc has drifted from the Eurasian continent, Taiwan has been fixed to the continent.

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The cusp appears to have been formed prior to the rotation of the south Ryukyu arc. When the south Ryukyu arc is restored to the palaeoposition according to the rotation angle estimated by Miki *et al.* (1990), the palaeo-Ryukyu and Taiwan–Luzon arcs meet at the junction with an included angle of 120° (Fig. 5). This angle is consistent with that observed at other cusps, e.g. Peru–Chile arcs junction, Kuril–northeast Japan arcs junction and southwest Japan–Ryukyu arcs junction. The reconstruction implies that the shape of the cusp had been formed by 10 Ma. The opening of the Okinawa Trough only decreased the included angle of the cusp from 120° to 100°.

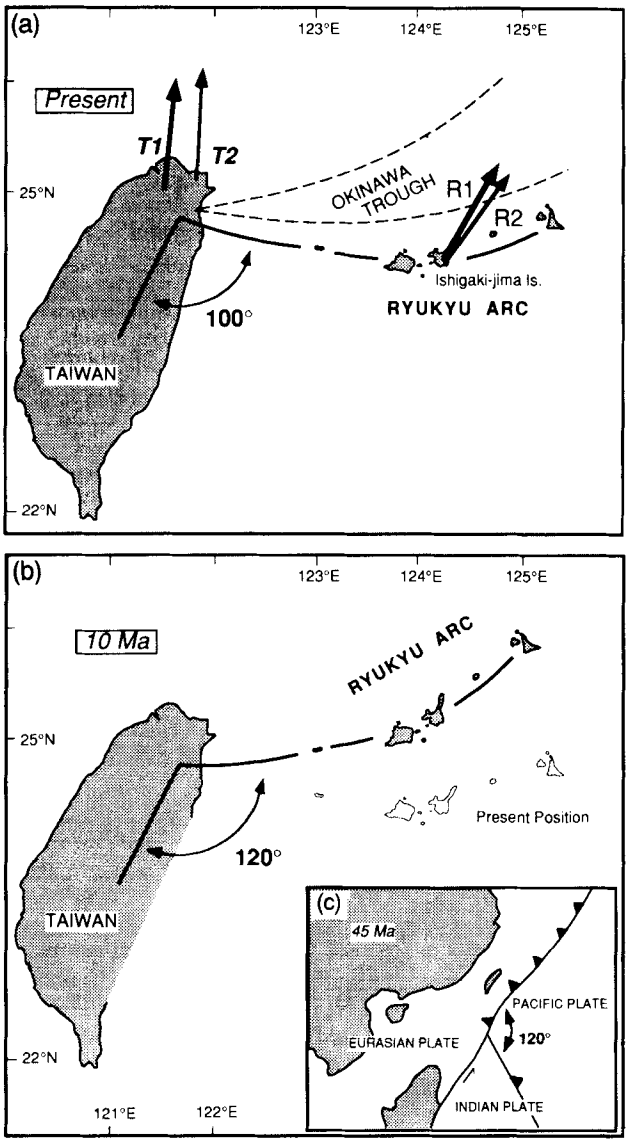


Figure 5. Cusp feature of the Ryukyu–Taiwan arc junction.
(a) Present feature of the Ryukyu–Taiwan arc junction modified from Miki *et al.* (1990). Arrows of *T*1 and *T*2 in northern Taiwan indicate the palaeomagnetic declinations of Miocene and Plio–Pleistocene, respectively. Arrows of *R*1 and *R*2 are palaeomagnetic declinations of Eocene and Miocene of the Ishigaki–Jima island in the south Ryukyu arc, respectively. Directions of thin arrows (*T*2 and *R*2) are from only one site.
(b) Palaeoposition of northern Taiwan (this study) and the south Ryukyu arc (Miki *et al.* 1990) at 10 Ma estimated from palaeomagnetic study. Neither rotation of the south Ryukyu arc nor the collision in Taiwan has been started until 10 Ma.
(c) Palaeoposition of triple junction of Eurasia, India and Pacific plates at 45 Ma (modified from Jolivet *et al.* 1989).

Collision of a buoyant ridge against a retreating trench is one of the possible mechanisms for the origin of a cusp (Vogt 1973). An analogue of the Vogt's model would be the collision of the Luzon arc with the Asian passive margin at the cusp of northern Taiwan. The cusp, however, was already formed before the collision, which began in late Miocene or Pliocene (e.g. Chi *et al.* 1981). The collision is therefore not responsible for the formation of the cusp in northern Taiwan.

The shape of some cusps along subduction zones appears to be an inherited feature of a plate boundary at the triple junction. At the triple junction, two of three plate boundaries form a cusp feature. A typical example is the junction of the Japan trench and the Izu–Bonin trench at the triple junction of the Pacific–Philippine–Eurasia plates. Northeast Japan and the Izu–Bonin arcs form an arc junction with the included angle of about 120°. If the third boundary becomes inactive (failed arm; Burke & Dewey 1973) or moves away, only the cusp feature will be observed. An example of cusp feature formed by failed arm is the Arica Bend in the Andes (Scanlan & Tunar 1992).

The cusp at the northern Taiwan may be a remnant feature of the plate boundary at the trench–trench–transform triple junction of the Pacific–Eurasia–Indian plates: the triple junction was situated at south of Taiwan in Eocene times (Jolivet, Huchon & Rangin 1989; Fig. 5), and moved southward to the present position of the north of Australia.

CONCLUSIONS

(1) Northern Taiwan has undergone neither tectonic rotation nor north–south translation with respect to the China block since 20 Ma.
(2) The arcuate shape of the geological structure in Taiwan is an inherited geometry rather than the result of secondary bending.
(3) While the south Ryukyu arc has drifted from the Eurasian continent during the last 10 myr, Taiwan has been fixed to the continent.
(4) The cusp at the Ryukyu–Taiwan arc junction had been formed by 10 Ma. Rotation of the Ryukyu arc only accentuates the cusp shape.

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APPENDIX

Petrographical descriptions of the samples for K-Ar dating

Site 24 (Miocene Chiopanshan volcanics): dolerite

A holocrystalline rock showing ophitic texture and consisting mainly of fresh idiomorphic plagioclase and fresh interstitial clinopyroxene. Fine-grained crystal aggregates which probably originally consisted of olivine and pyroxene also slightly occur.

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Site 43 (Miocene Chiopanshan volcanics): dolerite

A holocrystalline rock showing ophitic texture and consisting mainly of fresh idiomorphic plagioclase and fresh interstitial clinopyroxene. Fine-grained crystal aggregates also slightly occur. They are now pseudo-morphed but probably originally consisted of olivine and pyroxene. Small amounts of secondary muscovites are also locally present.

Site 40 (Pliocene Chilung volcano): porphyritic biotite hornblende dacite

The phenocrysts are of plagioclase, hornblende, biotite and a small amount of quartz, and are fresh. These are embedded in cryptocrystalline matrix.

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